Jon Morgan

Technical Director – Built Environment & Digital Integration

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"My work is centred on a balance of cross-disciplinary experience—spanning systems research, building performance simulation, and software automation—complemented by practical experience running a small consultancy. I focus on designing code-governed pipelines that connect physical building systems to sustainability reporting frameworks, ensuring automation is balanced with human oversight to translate operational metrics into reliable compliance outputs. This combination of hands-on technical experience and small business operations helps me act as a practical bridge, translating complex technical datasets into clear, transparent, and auditable structures for senior leadership."

Core Capabilities

- **Building Decarbonisation & Technical Transition Modelling:** Extensive experience in detailed simulation of energy, daylight, and building performance. Experienced in building the physical and financial models—integrating load curves, battery storage, solar sizing, and tariff analysis—that provide the analytical backing required for transition plans and help ground projections in physical data.
- **Building Performance & Multidisciplinary Engineering:** Ten years leading Arup's Australasian Building Physics team, specialising in building performance, energy systems, and operational efficiency. Experienced in coordinating complex building parameters across services (MEP), facades, fire safety, and acoustics, alongside mentoring and guiding engineering graduates.
- **Data Assurance & Automated Pipelines:** 30 years of programming experience (Python, SQL, Bash, Linux) used to architect automated, code-governed pipelines for calculations, combining automated speed with human-in-the-loop validation to minimise manual errors between physical building systems and ESG reporting platforms.
- **Strategic Operations & Governance:** Experience as a co-founder and director for nearly 8 years, managing operations, software deployments, and corporate governance. This operational background serves as a practical bridge, allowing me to align technical engineering delivery directly with corporate risk management, software infrastructure, and commercial property governance.

Professional Experience

Co-Perform Pty Ltd

2018 – Present

Co-Founder & Director

Co-founded and scaled this specialised built environment consultancy. As Director, I lead corporate rebranding, brand messaging, and commercial strategy while maintaining daily collaborative oversight of our operational pipelines, IT systems, and financial metrics.

- **Strategic Leadership & Commercial Oversight:** Drove our corporate rebranding, brand messaging, and growth strategy, maintaining close daily collaboration with my partner on firm-wide financials, cash position, proposals, invoicing, and work-in-progress (WIP).
- **Digital Operations & Process Control:** Exercised complete organisational control over our digital systems, managing internal IT support infrastructure and designing our core engineering calculation pipelines, automated processes, and project delivery systems.

- **Municipal Policy Strategy & Technical Evidence:** Developed the technical evidence base and policy strategy supporting the City of Melbourne Sustainable Building Policy, in partnership with Nation Partners. Focused on aligning municipal assets with Green Star pathways and identifying existing infrastructure to serve as extreme-heat cool centres.
- **Climate Resilience & Comfort Analysis:** Simulated building performance under projected future climate scenarios to stress-test designs, assessing thermal comfort and mitigating overheating risks during extreme heatwaves.
- **Embodied Carbon & Building Simulation:** Managed embodied carbon, daylight, and energy calculations for a major 6-star Green Star healthcare project. Developed a custom utility to parse contractor Bill of Quantities (BoQ) data, automating structural conversion for direct import into the carbon accounting platform.
- **Planning & Building Permit Support:** Delivered energy, thermal, and daylight simulations for Sustainability Management Plans (SMPs) to secure planning and building permits across industrial, commercial, residential, and community developments.
- **Systems Engineering & DevOps:** Built database-integrated pipelines using Python to automate compliance reporting and managed an AI-assisted DevOps repository to streamline cloud infrastructure and testing.
- **Disaster Recovery & Redesign:** Rapidly restored operations within 48 hours following a global hosting outage by rebuilding the corporate web presence using Astro.
- **Business Strategy:** Commissioned and co-led a six-month strategic review of the business model and market alignment to steer the firm's growth plans.

Moreland Energy Foundation Ltd & Visergy

2014 – 2018

Senior Consultant (MEFL) / Principal & Sole Trader (Visergy)

Delivered high-level energy efficiency engineering, regional decarbonisation models, and performance-based design simulations for municipal portfolios and commercial real estate.

- **Sole Trader Operations (Visergy):** Established and managed an independent consultancy, directing all operations, proposal development, project delivery, invoicing, and client relationships.
- **Solar Feasibility & Predictive Tooling:** Created a predictive modelling tool to right-size solar installations, accelerating early-stage design feasibility and concept planning.
- **Climate-Resilient Public Housing:** Collaborated with DesignInc on public housing prototypes engineered for future climate scenarios, analysing comfort and overheating risks to protect vulnerable occupants.
- **Portfolio Performance & Approvals:** Modelled energy, daylight, and thermal performance across diverse asset classes to support town planning, Sustainability Management Plans (SMPs), and public housing retrofit decisions.
- **Regional Emissions Modelling & Analysis:** Analysed cross-sectoral regional datasets to build emissions inventories for the Hepburn Shire Net-Zero transition plan, translating complex interval data into clear visual models for public stakeholders.

Arup (Melbourne)

2005 – 2014

Leader of the Australasian Building Physics Team

Directed regional analytical capacity, workflows, hardware integration, and detailed computational modelling for one of the world's premier engineering institutions.

- **Graduate Mentorship & Technical Training:** Mentored building performance graduates and delivered structured training for engineering rotations. Participated directly in regional graduate recruitment.
- **Interdisciplinary Performance Forecasting:** Coordinated complex design parameters across services, facades, and acoustics, using future climate scenarios in dynamic thermal models to forecast long-term performance.

- **Building Performance & Quality Assurance:** Standardised simulation workflows across offices in Australasia, ensuring technical rigour, auditable results, and quality control on dynamic compliance calculations.
- **Computational Platform & Tool Integration:** Maintained an internal compute platform to centralise calculations, writing custom integration scripts to link airflow and daylight simulations for complex designs.

Arup (UK) & BMT Fluid Mechanics

2002 – 2005

Consulting Engineer / Analyst

Delivered advanced airflow simulations, cladding pressure tests, facade load calculations, and process automation across commercial real estate, sports stadia, and offshore energy operations.

- **Built Environment Aerodynamics (BMT):** Used physical testing and simulation tools to conduct wind microclimate assessments, cladding pressure tests, and facade load calculations for complex tall buildings.
- **Systems & Modelling Automation:** Managed high-performance compute clusters and wrote custom 3D scripts to automate spatial modelling for design assessment tools.

Defence Science and Technology Organisation (DSTO)

1996 – 2000

Research & Development Systems Engineer

Executed advanced aerodynamic, hydrodynamic, and acoustic performance research for military aircraft, weapons systems, and submarines.

- **Aerospace & Defence Systems Engineering:** Conducted aerodynamic and hydrodynamic performance research for military aircraft (F/A-18 and F-111) and Collins-class submarines. Developed mathematical optimisation and fault-detection pipelines to diagnose jet engine degradation from physical test-bed logs.

Education & Accreditations

Professional Credentials & Industry Accreditations

- **Accredited Assessor — NABERS Upfront Carbon** (Embodied carbon assessment for materials, structure, and supply-chain metrics).
- **Energy & Sustainability Hub Advisory Board Member** — Building Designers Association of Victoria (2015 – 2018).

RMIT University

2009 – 2014

Master of Urban Planning and Environment

Advanced postgraduate studies focusing on urban environmental policy, microclimate resilience, sustainable infrastructure, and quantitative resource modelling.

RMIT University

1991 – 1995

Bachelor of Engineering (B.Eng.), Aerospace

Rigorous four-year engineering specialisation with deep focus on fluid dynamics and aerodynamics, structural mechanics, systems, control, and computational simulation.

Compliance & Operations

My approach is built on a practical balance: the broad, cross-disciplinary engineering background required to trace data across a portfolio's entire physical footprint, alongside a deep focus on the exact touchpoints where active building operations meet corporate compliance structures. Rather than prescribing a one-size-fits-all framework, I collaborate to map out precise technical connections—bridging utility metering setups, HVAC performance, and embodied carbon data directly into your corporate reporting systems.

For institutional property portfolios, navigating decarbonisation requires converting transition plans from high-level estimates into defensible capital investments. My experience in physical simulation—incorporating localized load curves, battery storage parameters, and tariff analysis models—helps me build reliable, adaptive models. By grounding transition strategy in verified, data-backed models, I help align technical plans with broader financial and corporate investment goals.

With a career that spans both technical engineering and business leadership, I view compliance and decarbonisation as integrated, operational challenges. I bring a balanced toolkit: the technical background to understand complex systems, paired with the practical experience of managing projects, budgets, and risk. This combination enables the design and configuration of automated, verifiable pipelines—grounded in experienced human-in-the-loop validation—that remove tracking risks, protect asset performance, and support consistent reporting integrity on the corporate balance sheet.

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